

Tracking Flames, Predicting Pressure, and Analyzing Jet Edges

An Automated Pipeline for Fluid and Combustion Dynamics

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Architecture & Pipeline Overview

This presentation consolidates three critical research pipelines developed to analyze macro-scale fluid and combustion behaviors.

- **Part 1: Automated Flame Tracking.** A highly robust pipeline for tracking flame propagation and extracting continuous mathematical flow dynamics.
- **Part 2: Pressure Dynamics Prediction.** Correlating heat release to pressure signals using deeply Stacked Vanilla Recurrent Networks.
- **Part 3: Spatiotemporal Segmentation.** Extracting sub-pixel fluid boundaries and droplet structures from 16-bit high-speed imaging.

Part 1

Automated Flame Tracking & Motion Analysis

Flame Tracking: Primary Objectives

Overarching Goal: To develop a fully automated pipeline for tracking flame propagation in high-noise experimental videos.

The Challenges:

- Rapid illumination variation causes global pixel intensity shifts, tricking background subtractors.
- Bright reflections on pipe walls create false "flames".
- Extracting velocity via direct numerical differentiation amplifies positional noise significantly.

The Solutions:

- **Computer Vision Phase:** Optimal background selection and Static Eraser boolean masking.
- **Mathematical Phase:** Saturating exponential curve fitting and analytical dynamics derivation.

Step 1: System Architecture

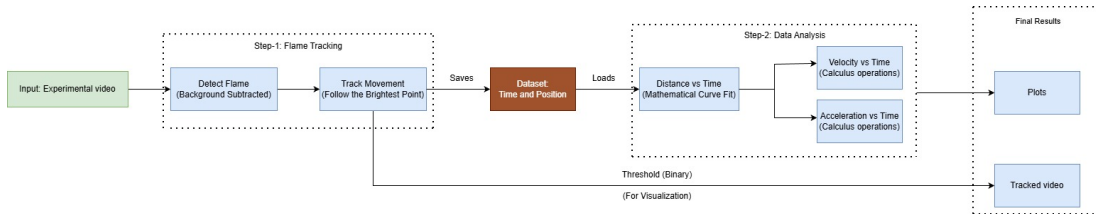


Fig: System architecture detailing the transition from raw experimental video to analytical plots.

Step 2: Region Definition & Background Selection

Spatial Zoning

- **Pipe Zone:** The primary bounding box containing the experiment.
- **ROI Masking:** A logical mask completely blacks out pixels outside the pipe, permanently preventing external lab motion from triggering the tracker.

Intelligent Background

- The algorithm checks the first 20 frames, cropping them strictly to the Pipe Zone.
- The frame with the absolute lowest total intensity is locked in as the Master Background, maximizing the signal-to-noise ratio.

Step 3: The Static Eraser Mask

Eliminating Pre-existing Spots

- Static white spots inside the geometry overlap with the dynamic flame front, dragging the tracker away.
- A threshold maps these problematic reflections, and a logical `bitwise_not` mathematically inverts them.
- This deletes static spots from live frames before tracking begins, ensuring an undisturbed environment.



Fig: Inherent white spots present inside the pipe geometry prior to ignition (Phi 1p2).



Fig: Red regions dictate where static spots are zeroed out.

Step 4: Morphology & HSV Enhancement

Targeted Cleaning

- Gaussian blur (5×5) and absolute difference subtraction isolate the moving foreground.
- A heavy 20×20 morphological OPEN kernel aggressively destroys floating noise, while a 3×3 CLOSE kernel heals the flame body.

Contrast Boosting

- The isolated flame is converted to HSV color space.
- The *Value* channel is scaled up, forcing the flame core to peak intensity while suppressing blurry fringes.

Step 5: Max Intensity Anchoring

Robust Target Lock

- Tracking the geometric centroid fails when the flame shape fractures or faces partial occlusion.
- By anchoring tracking logic strictly to the **Maximum Intensity** pixel, the system remains locked onto the hottest, densest core.
- `cv2.findContours` then shrink-wraps the bounding box tightly to the active flame shape.

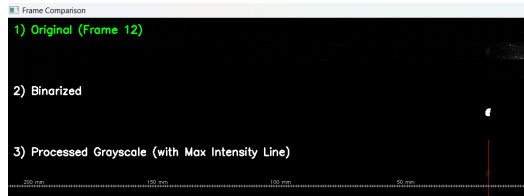


Fig: Tracker anchors precisely on max intensity.

Step 6: Temporal Normalization

Data Standardization

- High-speed cameras operate at varied frame rates, making direct comparisons difficult.
- To allow for 1:1 cross-experiment comparison, raw timestamp arrays are multiplied by a standard scaling factor.
- This mathematically compresses the dataset into a strict 5.0-second analytical window without altering the physical shape of the data.

Normalization Formula

$$\text{Scaling Factor} = \frac{\text{TARGET_DURATION (5.0s)}}{\text{Original Max Time}}$$

Step 7: Saturating Exponential Modeling

Curve Fitting

- We fit a mathematical model to represent the fundamental physical progression of a decelerating flame front.
- Scipy's `curve_fit` optimizes parameters, locking A to the empirical saturation boundary (average of the last 5 frames).

Position Model $P(t)$

$$P_{eq}(t) = A \left(1 - e^{-k(t-t_0)} \right) + C$$

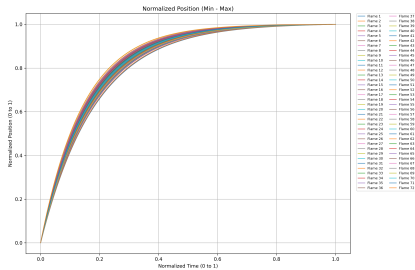


Fig: Normalized position bounded by exponential fit.

Step 8: Analytical Motion Dynamics (Velocity)

Smooth Extraction

- Derived analytically directly from the continuous position model.
- Yields perfectly smooth profiles completely devoid of high-frequency experimental jitter.
- Demonstrates peak velocity at ignition followed by a continuous exponential decay.

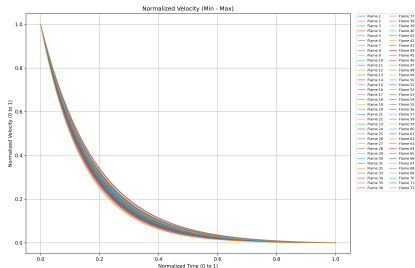


Fig: Analytically Derived Velocity Profile.

Velocity Profile $V(t)$

$$V_{eq}(t) = Ak e^{-k(t-t_0)}$$

Step 9: Analytical Motion Dynamics (Acceleration)

Resistance Profiling

- The second derivative reveals a smooth, purely negative acceleration curve.
- This mathematically confirms the flame front experiences maximum resistance forces immediately after initial explosive expansion.

Acceleration Profile $A_{cc}(t)$

$$A_{eq}(t) = -Ak^2 e^{-k(t-t_0)}$$

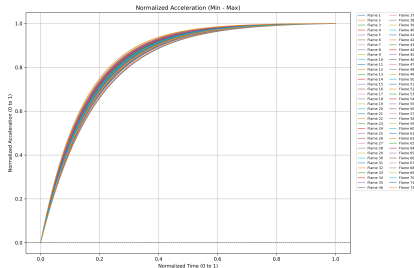
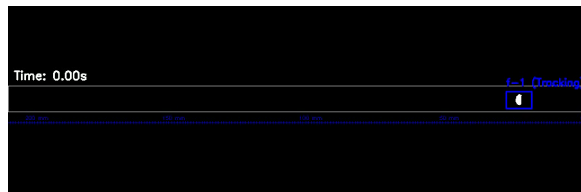


Fig: Analytically Derived Acceleration Profile.

Flame Tracking: Validation & Outputs

System Verification

- The tracker reliably outputs pristine (t, P_{mm}) CSV datasets without centroid skewing or artifact jumping.
- Visual tracking securely maintains anchor on maximum intensity core throughout the entire run.
- The non-dimensional "Star" transform accurately generalizes dynamics across different equivalence ratios.



*Fig: Processed tracking output
(Phi_1p0_u_0p3_C001H001S0001). (Click image to view
video).*

Part 2

Predictive Sequence Modeling

Pressure Dynamics: Predictive Challenges

Objective: Predict continuous pressure signals from unseen heat input signals using recurrent networks.

Modeling Obstacles

- **Data Volume:** Excessive raw sequence points from high-frequency sensors create massive training overhead.
- **Cold-Start Transients:** Neural network hidden states initialized at zero cause severe prediction lag at the start of a window.
- **Sequence Memorization:** Deep networks risk memorizing chronologies rather than learning continuous physical manifolds.

Step 1: Temporal Downsampling

Dual-Purpose Decimation ($\text{STEP_SIZE} = 2$)

- Processing every single micro-second leads to redundant calculations. We apply a uniform temporal decimation strategy.
- **Computational Focus:** Actively reduces dataset size by 50%, drastically shortening sequence length for Backpropagation Through Time (BPTT).
- **Physical Focus:** Acts as a low-pass filter, stripping non-physical high-frequency artifacts to prevent the model from fitting sensor noise.

Step 2: Resolving Cold-Start Transients

The Initialization Problem

- By default, RNNs initialize their internal memory (h_0) to absolute zero.
- If a target prediction window starts at a moment of extreme high pressure, the network is blind to system momentum, resulting in massive transient errors.
- **Solution (Warm-Up Buffer):** The dataset class extracts a 2000-step slice. The network pre-processes the first 500 steps strictly to populate hidden states (h_t) before any loss is calculated, physically aligning its momentum.

Step 3: Trajectory Feature Engineering

The 2D Input Tensor

- Feeding only current heat (H_t) forces the network to blindly guess the physical trajectory.
- We explicitly engineer the discrete derivative ($\Delta H_t = H_t - H_{t-1}$) to provide wave acceleration direction.
- Signals are processed via Z-Score normalization to prevent activation saturation while preserving acoustic variance.

Engineered Tensor

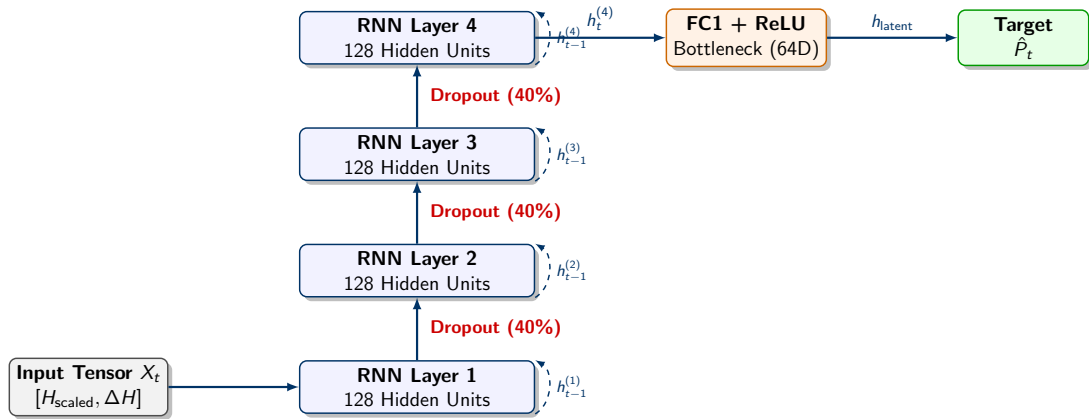
$$X_t = \begin{bmatrix} H_{scaled}, & \Delta H \end{bmatrix}$$

Step 4: Deep Stacked Architecture

4-Layer Vanilla RNN

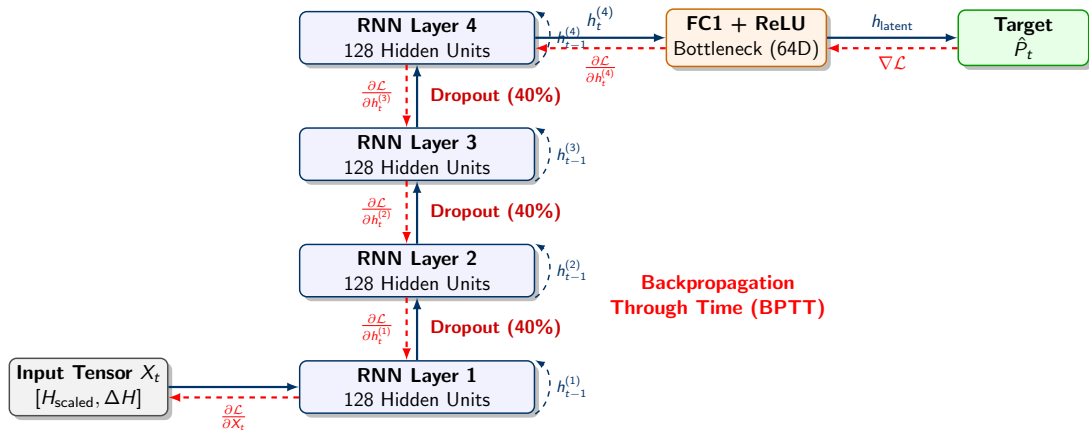
- Because our feature engineering explicitly encodes trajectory, complex LSTM gating mechanisms proved unnecessary.
- We rely on a deeply stacked, 4-layer Elman Recurrent Network featuring 128 hidden units per layer.
- This depth allows the model to organically combine basic gradients into complex macroscopic wave structures.
- The 128D output is passed through an Information Bottleneck (64D) to strip final noise before projecting to a 1D Pressure scalar.

Information Flow: Forward Pass & Backpropagation



Step 1: Forward Pass & Recurrent State Updates. Input data and previous hidden states (h_{t-1}) are processed simultaneously to predict pressure dynamics.

Information Flow: Forward Pass & Backpropagation



Step 2: Backpropagation Through Time. Gradients ($\nabla \mathcal{L}$) flow backwards to update weights, calculating partial derivatives at each depth.

Step 5: Architectural Regularization

Preventing Sequence Memorization

- **Heavy Dropout (40%):** Applied between every stacked layer. This actively shatters neuron co-adaptation, forcing independent feature learning rather than rote memorization.
- **Stochastic Windowing:** Batches are built by randomly slicing the dataset rather than reading it chronologically.
- Shuffling start indices completely breaks chronological sequence learning, forcing the model to map localized dynamics.

Time Domain Results

- The model mapped 1500+ steps of unseen thermal inputs to pressure dynamics in a strictly open-loop regime.
- Near-perfect Pearson Correlation confirms the Warm-Up phase successfully eliminated initialization phase lag.
- **Metrics:** Avg RMSE: 0.0063, Avg R^2 : 0.9857, Avg PCC: 0.9934.

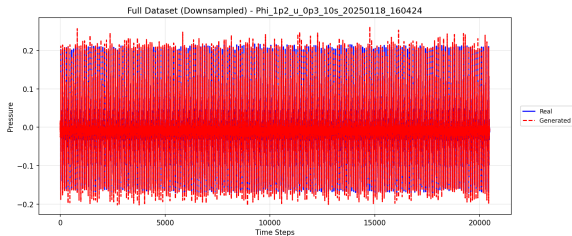


Fig: Continuous generative pressure output vs empirical ground truth.

Validation in the Time Domain: Uniform Window Sampling

Evaluating Generalization Across the Sequence

To rigorously evaluate the model's stability, we extracted six uniformly spaced inference windows (W_1 to W_6) across the entire unseen dataset using a linear space distribution. Each window represents 1500 time steps of continuous open-loop prediction (strictly evaluated after the 500-step warm-up phase). This confirms that generative accuracy is consistent and independent of specific starting regimes.

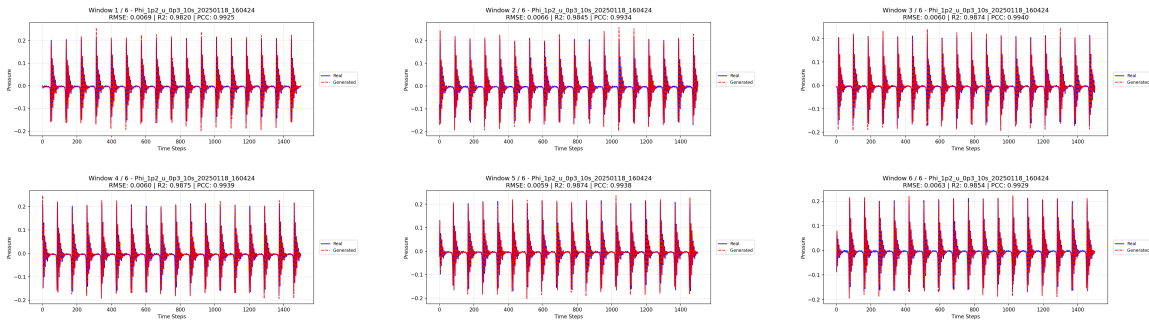


Fig: 1500-step pressure predictions across six distinct temporal locations.

Topological Validation (Phase Space)

Phase Space Isomorphism

- We compared the physical attractor (P vs ΔP) to latent projections to verify if the network learned the true system topology.
- The RNN's 128D internal memory organically projects into a smooth, bounded 2D envelope.
- This envelope perfectly mirrors the physical ground truth limit cycle.

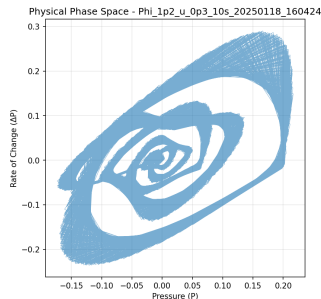


Fig: Physical Phase Space (P vs ΔP).

Latent Space Projection

Validation of Learning

- This structural similarity proves the model has not merely memorized sequential data points.
- The absence of fractured trajectories guarantees learned dynamics remain physically stable over long, continuous generation windows.

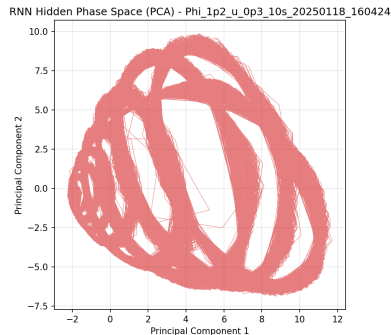


Fig: Hidden Phase Space (PCA projection).

Temporal Determinism (Takens' Theorem)

3D Embedding Analysis

- Applying Takens' Embedding Theorem to sequential time delays $(h(t), h(t+1), h(t+2))$ unfolds into a continuous 3D structure.
- State transitions are gradual and cohesive, showing a clear central void characteristic of periodic oscillatory motion.
- Strictly rules out chaotic mode-hopping or exploding gradients within the vanilla RNN.

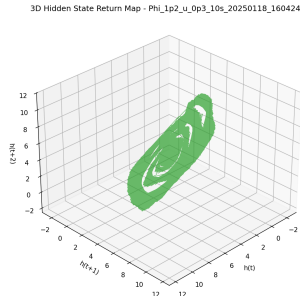


Fig: 3D Hidden State Return Map.

Summary of Findings

- **Architectural Efficiency:** A deeply stacked (4-layer) Vanilla RNN effectively outperforms complex gated LSTM networks by mapping sequential dynamics via explicit feature derivatives.
- **Initialization Stability:** Implementing a 500-step Warm-up Buffer successfully stabilized hidden state matrices, completely resolving initial phase errors.
- **Topology:** The model proved capable of mapping highly stable continuous limit cycles, operating purely on learned physical rules rather than noise.

Part 3

Unsupervised Spatiotemporal Segmentation

Segmentation: Core Objectives

Overarching Goal: Ultra-precise, sub-pixel edge tracking of fluid structures from 16-bit high-speed imaging.

The Challenges:

- Low-contrast mist and specular reflections obscure true boundaries.
- Standard global binarization suffers from severe pixel-density bias, causing over-segmentation.
- Identifying the true jet structure is hindered by disconnected optical noise and internal hollow voids.

Step 1: 16-Bit Local Range Inversion

Dimensional Stability

- Applies a strict 16-bit range inversion strictly relative to the localized physical peak of the current frame.
- This mathematically prevents global exposure drift and ensures dimensional stability across frames before processing.

Formula

$$I_{inv}^t(i, j) = \frac{I_{max}^t - I_t(i, j)}{I_{max}^t - I_{min}^t + 10^{-6}} \times 65535$$

Step 2: Local Min-Max Inversion
 $((I_{max} - I_t) / (I_{max} - I_{min})) * 65535$

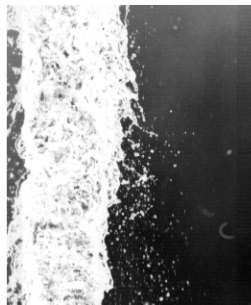


Fig: Local Min-Max Inverted Frame.

Step 2: Dynamic Nested GMM Processing

Statistical Shape Extraction

- Executes an unsupervised statistical breakdown of the frame's intensity spectrum using Gaussian Mixture Models (GMM).
- Spreads four initial distributions evenly across the intensity range.
- Forces the algorithm to acknowledge all physical density states without bias.

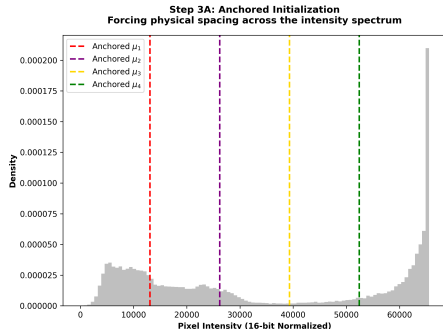


Fig: Intensity initialization spread forcing physical spacing.

Step 3: Expectation-Maximization & Fringe Isolation

Cluster Identification

- Calculates membership probability (γ) and adjusts expected centers (μ) to perfectly fit assigned pixels.
- Sorts distributions from darkest to brightest, isolating the 3rd cluster (C_2).
- This isolates the critical physical transition zone (fringe) between mist and dense fluid.

Step 3D: 2D GMM Hard Assignment Map
(Mapped on Test Frame 150)

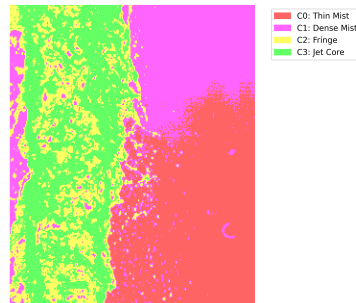


Fig: 2D GMM Hard Assignment Map isolating the fringe.

Step 4: Hierarchical Threshold Anchoring

Target Binarization

- Applies a secondary, Hierarchical 2-component GMM exclusively onto the isolated C_2 cluster to split the transition zone into a lighter "sub-mist" and a denser "sub-core".
- A structural offset (δ_{shift}) anchors the strict binarization threshold.
- Evaluates every spatial pixel against this dynamically generated cutoff.



Fig: Dynamically Binarized Target derived from threshold anchoring.

Step 5: Priority Graph Search (CCL)

Physical Constraint Prioritization

- Extracts components using 8-connectivity graph-based search.
- The component set is searched iteratively to identify the largest component anchored at the source boundary ($Y = 0$).
- Automatically bypasses disconnected ambient noise clusters.



Fig: Max Area component isolated via physical constraint.

Step 6: Consolidation & Capping

Noise Mapping & Structural Sealing

- Remaining fragmented clusters are mathematically united into a collective 'Noise Map'.
- Extracts extreme X-coordinates exclusively at the absolute top and bottom Y-rows of the priority mass.
- Creates a rigid mathematical cap strictly within the object's physical bounds, ensuring perfect structural sealing.

Step 6: Horizontal Capping

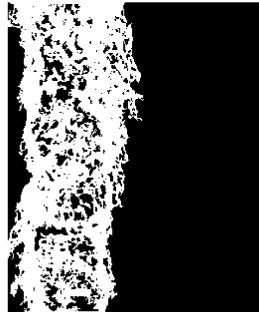


Fig: Row-specific horizontal capping.

Step 7: Topological Contour Closure

Resolving Internal Voids

- Central specular reflections cause hollow voids inside the fluid boundary.
- The pipeline invokes a topological flood-fill operation (Jordan Curve Theorem) to define the solid core as the union of the boundary and its bounded topological interior.
- Converts the incomplete structure into a completely solid binary mask.



Fig: Flood-filled contour closure converting to a solid binary mask.

Step 8: Morphological Isolation

Detaching Artifacts

- Applies a morphological opening operation using an elliptical structuring element (3×3).
- Severs weak, thin fluid filaments connecting noise to the main body.
- 8-way CCL is re-applied, partitioning clusters by descending area to definitively extract the geometric fluid core.



Fig: Final topological core isolation.

Step 9: Ghost Fragment Verification

Temporal Interrogation

- Detached fragments are isolated by computing spatial deviation between the filled mask and the core.
- Candidates are interrogated against historical search tracks.
- Fragments intersecting previous boundaries are permanently verified as "Ghost" noise and deleted via Boolean NOT algebra.

Step 10: Ghost Verification

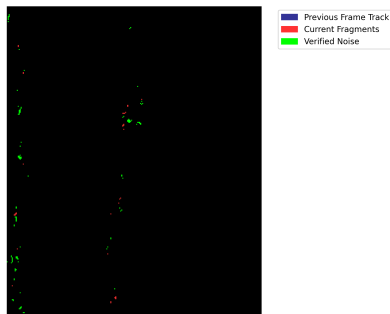


Fig: Verified ghost noise highlighted in green.

Step 10: Topological Boundary Extraction

Continuous Contour Tracing

- Traces the continuous outer contour of the verified mask.
- Identifies the absolute right-most anchor points at top and bottom bounds, splitting the contour path.
- Retains exclusively the right-most topological edge for accurate curvature analytics.

Step 13: Topological Boundary Extraction

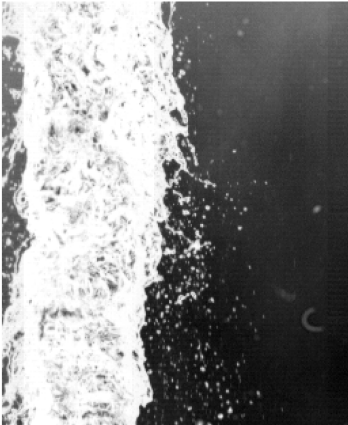


Fig: Rightmost boundary extracted.

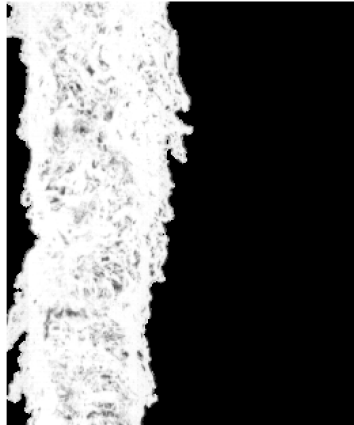
Final Result & Validation

- Comparison between Normalized raw image and Predicted jet of "10kV ROI2 FV 75kfps 1p7slpm R1 C001H001S0001".

Step 2: Local Min-Max Inversion



Step 14: Final Grayscale Target Extracted



Pristine Output & IoU

- The finalized mask is mapped directly onto the locally inverted original data, nullifying background while preserving 16-bit intensities.
- Compared against manual annotation by calculating Intersection over Union (IoU).
- Achieved a 98% overlap score, proving high algorithmic robustness.

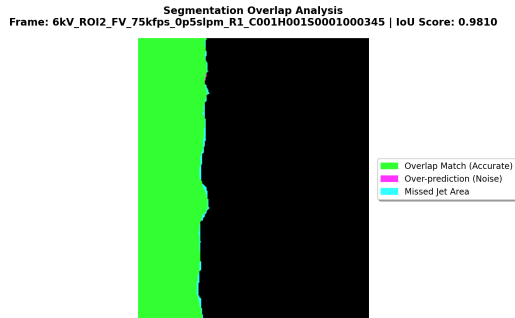


Fig: 98% Segmentation IoU overlap.

- **Flame Tracking:** We built an automated pipeline resilient to extreme glare, anchoring strictly to maximum intensity to flawlessly model saturating exponential velocity dynamics.
- **Sequence Prediction:** A compact 4-layer Stacked RNN successfully utilized trajectory derivatives to bypass memorization, generating high-accuracy 3D phase spaces from raw heat data.
- **Spatiotemporal Segmentation:** Applying localized GMMs and robust Jordan Curve closure effectively bypassed pixel-density bias, delivering pristine 98% physical tracking accuracy in severe environments.

Thank you! Questions?